

Tarea 3.5

$$5) x^2 - 4xy + y^2 = 4$$

$$2x - 4(xy' + y) + 2yy' = 0$$

$$2x - 4xy' - 4y + 2yy' = 0$$

$$-4xy' + 2yy' = -2x + 4y$$

$$y' = \frac{-2x + 4y}{-4x + 2y} = \frac{-2(x + 2y)}{2(2x + y)} = \frac{-x + 2y}{2x + y}$$

$$9) \frac{x^2}{x + y} = y^2 + y$$

$$\frac{2xy'(x+y) - x^2(y'+1)}{(x+y)^2} = 2y$$

$$\frac{2x^2y' + 2xy'y - x^2y' - x^2}{(x+y)^2} = 2y$$

$$\frac{x^2y' + 2xy'y - x^2}{(x+y)^2} = 2y$$

$$x^2y' + 2xy'y - x^2 = 2y(x+y)^2$$

$$x^2y' + 2xy'y - x^2 = 2y(x^2 + 2xy + y^2)$$

$$x^2y' + 2xy'y - x^2 = 2x^2y + 4xy^2 + 2y^3$$

$$x^2y' + 2xy'y = 2x^2y + 4xy^2 + 2y^3 + x^2$$

$$y' = \frac{2x^2y + 4xy^2 + 2y^3 + x^2}{x^2 + 2x}$$

$$11) x^2 y^2 + x \sin y = 4$$

$$x^2 2y y' + y^2 2x + x \cos(y) y' + \sin y = 0$$

$$x^2 2y y' + x \cos y y' = -y^2 2x + \sin y$$

$$y' = \frac{-2xy^2 - \sin y}{2x^2 y + x \cos y}$$

$$15) e^{xy} = x - y$$

$$e^{xy} \left(\frac{y(1) - xy'}{y^2} \right) = 1 - y'$$

$$y^2 y' - xy' e^{xy} = y^2 - y e^{xy}$$

$$y' = \frac{y^2 - y e^{xy}}{y^2 - x e^{xy}}$$

$$y e^{xy} - x y' e^{xy} = y^2 - y^2 y'$$

$$17) \tan^{-1}(x^2 y) = x + x y^2$$

$$\frac{1}{1 + (x^2 y)^2} \star x^2 y' + y 2x = x(2y) y' + y^2$$

$$x^2 y' + 2xy = x(2y) y' + y^2 (x^2 y)$$

$$x^2 y' - 2xy y' = x^2 y^3 - 2xy$$

$$y' = \frac{x^2 y^3 - 2xy}{x^2 - 2xy}$$

$$19) \sin(xy) = \cos(x+y)$$

$$\cos(xy) (x y' + y) = -\sin(x+y) (1+y')$$

$$\cos(xy) x y' + \cos(xy) y = -\sin(x+y) - \sin(x+y) y'$$

$$\cos(xy) x y' + \sin(x+y) = -\sin(x+y) - \cos(xy) y$$

$$y' = \frac{-\sin(x+y) + \cos(xy)}{x \cos(xy) + \sin(x+y)}$$

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$$23) x^4 y^2 - x^3 y + 2xy^3 = 0$$

$$x^4 2y' + y^2 4x^3 x' - x^3 + y 3x^2 x' + 2(x 3y^2 + y^3 x') = 0$$

$$2x^4 y' + 4x^3 y^2 x' - x^3 + 3x^2 y x' + 6xy^2 + 2y^3 x' = 0$$

$$4x^3 y^2 x' + 3x^2 y x' + 2y^3 x' = -2x^4 y' + x^3 + 6xy^2$$

$$y' = \frac{-2x^4 y' + x^3 - 6xy^2}{4x^3 y^2 - 3x^2 y + 2y^3}$$

$$27) x^2 - xy - y^2 = 1 \quad (2, 1) \quad (\text{hipérbola})$$

$$2x - (xy' + y) - 2yy' = 0$$

$$-xy' - 2yy' = -2x + y$$

$$y' | (2, 1) = \frac{2(2) - 1}{2 + 2(1)} = \frac{3}{4}$$

$$y' = \frac{2x - y}{x + 2y}$$

$$y - 1 = \frac{3}{4}(x - 2)$$

$$y = \frac{3}{4}x - \frac{1}{2}$$

$$31) 2(x^2 + y^2)^2 = 25(x^2 - y^2) \quad (3, 1) \quad (\text{lemniscata})$$

$$2[2(x^2 + y^2)(2x + 2yy')] = 25[2x - 2yy']$$

$$4(x^2 + y^2)(2x + 2yy') = 25(2x - 2yy')$$

$$4(3^2 + 1^2)(2(3) + 2(1)y') = 25(2(3) - 2(1)y')$$

$$240 + 80y' = 750 - 50y'$$

$$80y' + 50y' = 750 - 240$$

$$y' = \frac{-90}{130} = -\frac{9}{13}$$

$$y - 1 = -\frac{9}{13}(x - 3)$$

$$y - 1 = -\frac{9}{13}x + \frac{27}{13} + 1$$

$$y = -\frac{9}{13}x + \frac{40}{13}$$

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$$37) \sin y + \cos x = 1$$

$$\cos y (y') - \sin x = 0$$

$$\cos y (y') = \sin x$$

$$y' = \frac{\sin x}{\cos y}$$

$$y'' = \frac{\cos x \cos y - \sin x (-\sin y y')}{(\cos y)^2}$$

$$y'' = \frac{\cos x \cos y + \sin x \sin y y'}{(\cos y)^2}$$

$$y'' = \frac{\cos x \cos y + \sin x \sin y \left(\frac{\sin x}{\sin y}\right)}{(\cos y)^2}$$

$$y'' = \frac{\cos^2 y \cos x + \sin^2 x \sin y}{\cos^3 y}$$

45) Determine una ecuación para la recta tangente a la hipérbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2yy'}{b^2} = 0$$

$$\frac{b^2 x}{a^2 y} = y' \quad (2)$$

$$y - y_0 = \frac{b^2 x_0}{a^2 y_0} (x - x_0) \quad \frac{2x}{a^2} = \frac{2yy'}{b^2}$$

$$\frac{b^2 x_0}{a^2 y_0} = m$$

$$y = \frac{b^2 x_0}{a^2 y_0} x - \frac{b^2 x_0^2}{a^2 y_0} + y_0$$

$$51) y = \sin^{-1}(2x+1)$$

$$y' = \frac{1}{\sqrt{1-(2x+1)^2}} (2) = \frac{1}{\sqrt{1-x^2-x}}$$

$$57) y = x \sin^{-1} x + \sqrt{1-x^2}$$

$$y' = x \left(\frac{1}{\sqrt{1-x^2}} \right) + \sin^{-1} x + \frac{1}{2\sqrt{1-x^2}} (-2x) = \sin^{-1} x + x \frac{1}{\sqrt{1-x^2}} + \frac{1}{2\sqrt{1-x^2}} (-2x)$$

$$y' = \sin^{-1} x$$

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$$+5) \quad x^2 y^2 + xy = 2$$

$$x^2(2yy') + 2xy^2 + xy' + xy' + y = 0$$

$$2x^2yy' + xy' = -y - 2xy^2$$

$$y' = -\frac{y + 2xy^2}{2x^2y + x} = -1$$

$$-(y + 2xy^2) = -(2x^2y + x)$$

$$y + 2xy^2 = 2x^2y + x$$

$$y = x$$

$$x^2x^2 + x \cdot x = 2$$

$$x^4 + x^2 = 2 = 0$$

$$(x^2 + 2)(x^2 - 1) = 0$$

$$x^2 + 2 = 0$$

$$(x-1)(x+1) = 0$$

$$x = \pm 1$$

$$y(1 + 2xy) = x(2xy + 1)$$

$$y = x$$

$$\underline{(-1, -1), (1, 1)}$$